



Spark Team

أسئلة سنوات سابقة لمادة

الالكترونيات 1

شكوت إلى وكيع سوء حفظي قأرشدني إلى ترك المعاصي
وأخبرني بأن العلم نور ونور الله لا يؤتي لعاصي



Q-1- For the circuit in fig.1, answer the following three questions:

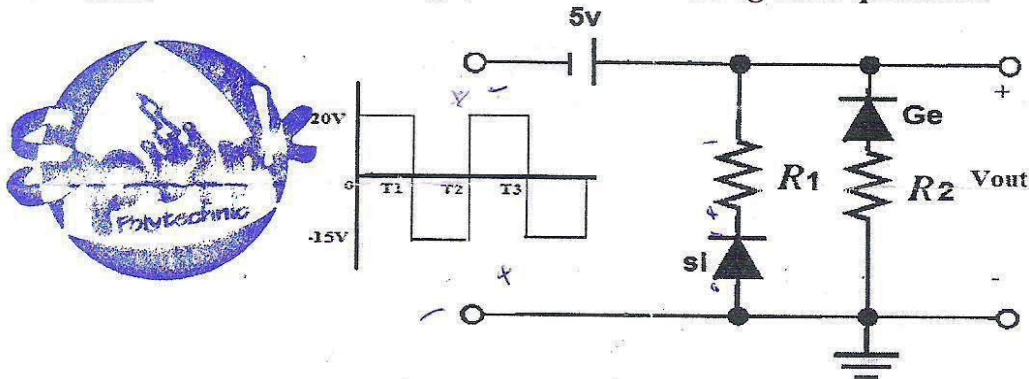


Fig1

- The output voltage at the positive half cycle is: 25 Volt
- The output voltage at the negative half cycle is: -10 Volt
- The voltage across R1 at the negative half cycle is: Zero
- The voltage across R2 at the negative half cycle is: +9.7 Volt

Q.2- for the circuit shown in fig.2, answer the following questions:

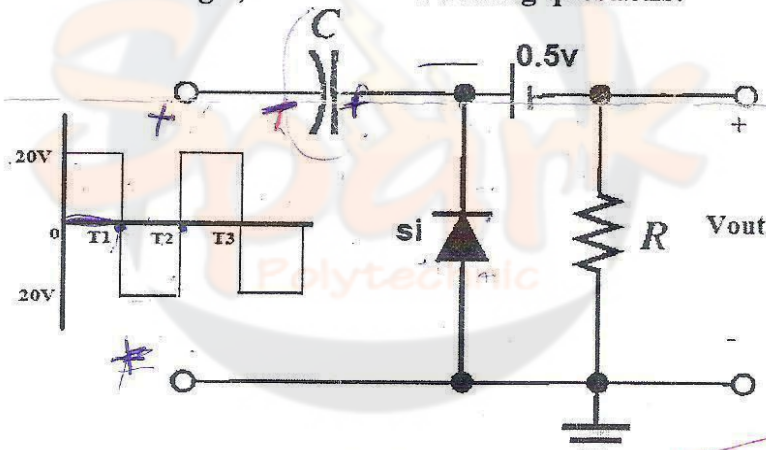


Fig2

- The value that the capacitor will be charged is: 19.3 Volt
- The output voltage at the period (T1 to T2) is: -1.2 Volt
- The output voltage at the period (T2 to T3) is: 38.8 Volt

Q.3- for a full wave bridge with germanium diodes, if the sinusoidal input peak is 50V and the input frequency is 20HZ then answer the following:

- The peak sinusoidal output is: 49.4 Volt
- The average output voltage (V_{DC}) is: 31.4184 Volt
- The output frequency is: 40 Hz



spark team

Al-Balqa' Applied University



جامعة البلقاء التطبيقية

Electronic devices

Second exam 3 /12/2012

Q-1- For the voltage regulator circuit in fig.1, answer the following questions:

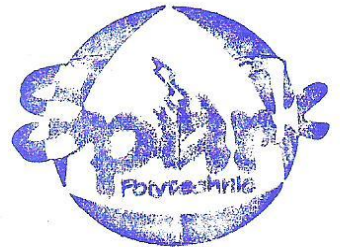
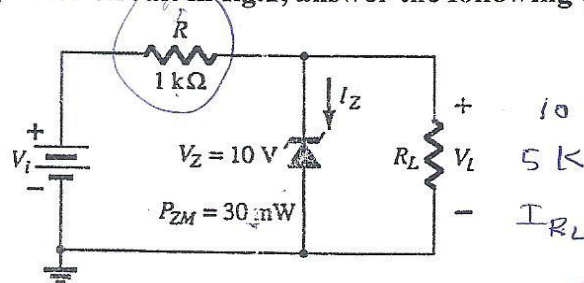


Fig1

A. If $R_L = 5 k\Omega$ then the maximum input voltage is: 15 volt

B. If $V_i = 30V$ then the minimum load resistance is: 500 Ω

Q.2- Answer the following about figure 2 (the zener voltage at forward bias is 0.7V):

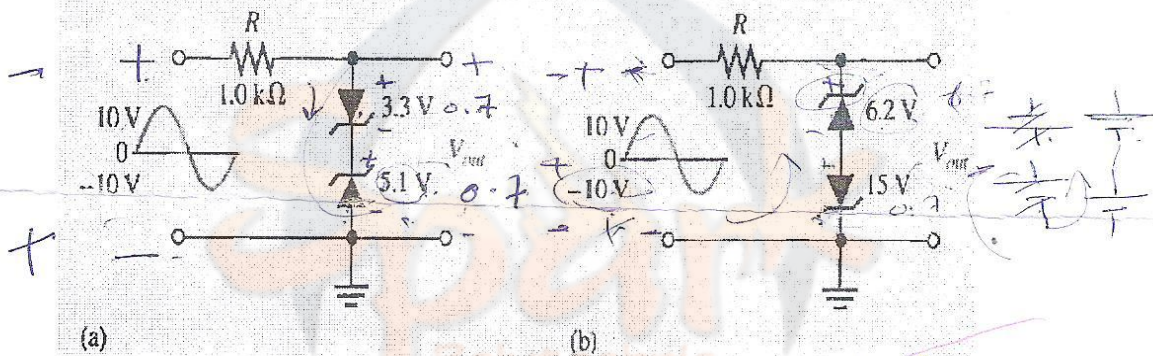


Fig2

A. The maximum output on figure (a) at positive half cycle is: 5.8 V

B. The maximum output on figure (a) at negative half cycle is: -4 V

C. The maximum output on figure (b) at positive half cycle is: 10 V (16 ϕ)

D. The maximum output on figure (b) at negative half cycle is: -10 V

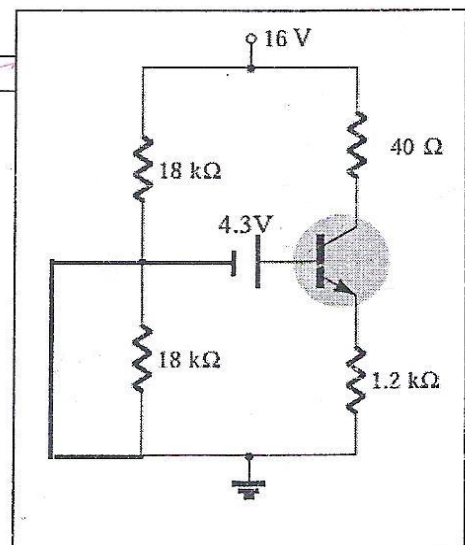
Q.3- for the circuit shown below, answer the following questions:

A. The value of R_{TH} seen from the base is: Zero

B. The value of V_{TH} seen from the base is: 4.3 V

C. The value of I_C is: 3 mA

D. The value of V_{CE} is: 12.28





Electronic devices

First exam 8/4/2010

Q-1- Determine V_o and I_D for the network of Fig. 1.

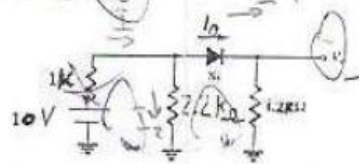


Fig. 1

Q-2-(a) Given $P_{max} = 14 \text{ mW}$ for each diode of Fig. 2. determine the maximum current rating of each diode (using the approximate/equivalent model).

(b) Determine I_{max} for $V_{max} = 160 \text{ V}$.

(c) Determine the current through each diode at V_{max} using the results of part (b).

(e) If only one diode were present, determine the diode current and compare it to the maximum rating.

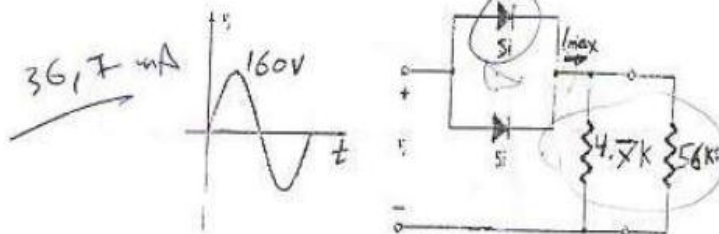


Fig. 2

Q-3- Sketch v_o for the network of Fig. 3. and determine the dc voltage available.

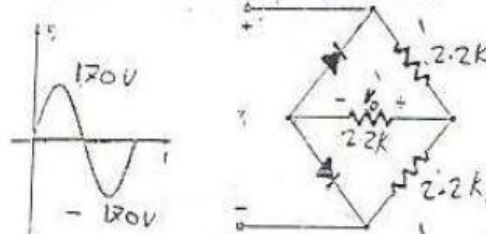


Fig. 3

Q-4- Sketch v_o for the network of Fig. 4 for the input shown.

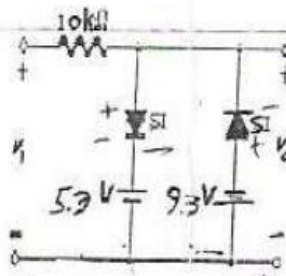
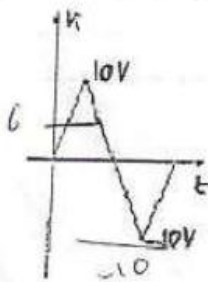


Fig. 4



Q.1

(9 marks)

Multiple Choices from 1-11

1) What is the state of an ideal diode in the region of nonconduction?

- a) An open circuit b) A short circuit
d) Unpredictable e) Undefined

2) The diode _____.

- a) is the simplest of semiconductor devices
b) has characteristics that closely match those of a simple switch
c) is a two-terminal device
d) All of the above

3) The diffused impurities with _____ valence electrons are called donor atoms.

- a) 4 b) 3 c) 5 d) 0

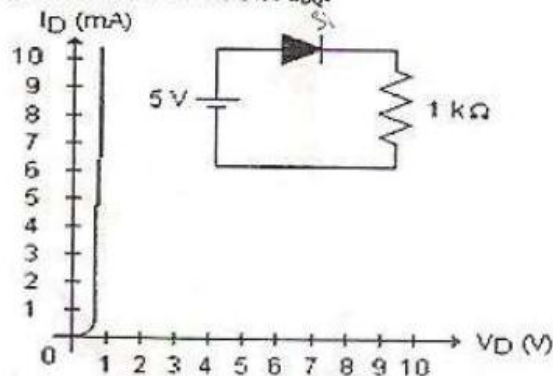
4) Calculate static resistance R_D of a diode having $I_D = 30 \text{ mA}$ and $V_D = 0.75 \text{ V}$.

- a) 25Ω b) 40Ω
c) 0.04Ω d) 0.025Ω

5) Any electron that has left its parent atom has _____ energy state relative to any electron in the atomic structure.

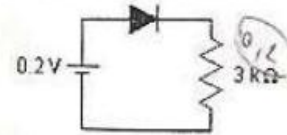
- a) The same b) a lower
c) A higher d) an undefined

6) Use the information provided here to determine the value of I_{DQ} .



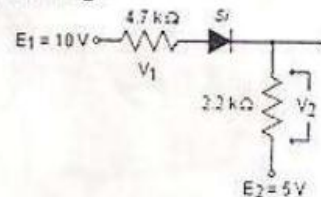
- a) 0mA b) 4.1mA
c) 5mA d) 10mA

7) Determine the voltage across the resistor.



- a) 0V b) 0.09V c) 0.2 d) 0.44V

8) Determine V_2 .

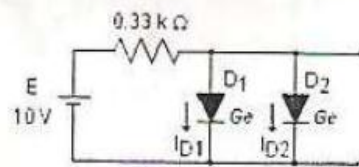


- a) 3.201V b) 0V c) 4.3V d) 1.371V

9) A short circuit has a _____ drop across its terminals, and the current is limited only by the surrounding network.

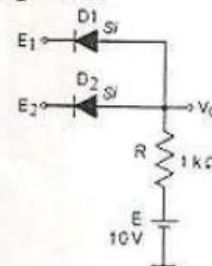
- a) 5V b) 0V c) 1V d) ∞

10) Determine I_{D2} .



- a) 29.40 mA b) 30.30 mA
c) 14.7 mA d) None of the above

11) Determine the current through each diode if $E_1 = E_2 = 0 \text{ V}$.



- a) 4.65 mA b) 9.3 mA
c) 18.6 mA d) 0.7 mA



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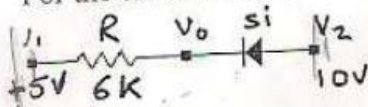
First exam
Electronic Device

Dr. Hasan Al-Harasis
Eng. Kawla Omar

19

1) Determine the level of V_o and I_D . (4 points)

For the circuit below



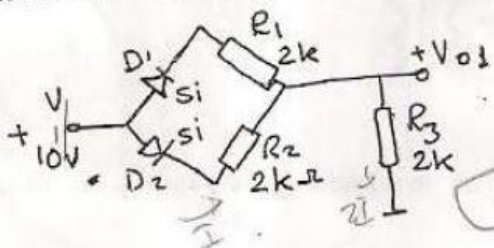
$$V_1 - IR - V_o + V_D - V_2 = 0$$

$$V_o = 5 - 6IR + 0.7 - 10$$

$$V_o = 10 - 5.3 = 4.3V \quad IR = \frac{8.6}{6K}$$

Determine the level of V_o , I_D , V_{dc} , PIV . (4 points)

$$I_D = \frac{8.6}{6K} = 1.43mA$$



$$-10 + 0.7 + 2I + 2(2K) = 0$$

$$V_o = 6.12$$

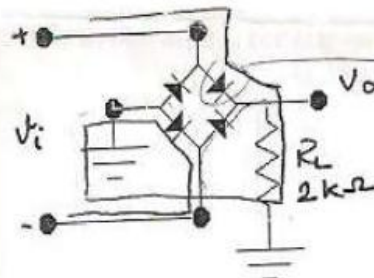
$$I = 1.55mA$$

2) In the circuit shown in fig3, assume ideal diodes if $v_i(t) = 100 \sin \omega t$ (6 points)

Draw V_o

calculate the DC output voltage.

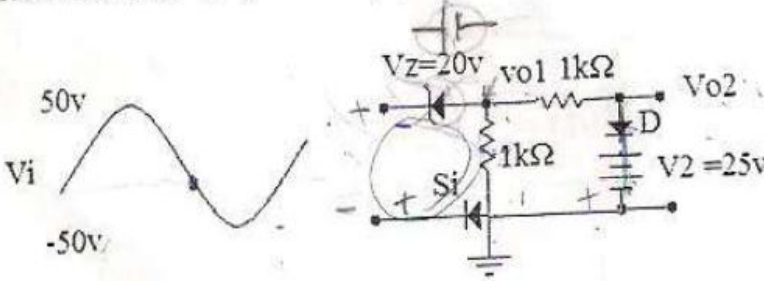
Determine the PIV for each diode.



$$100(0.636)$$

$$0.636 \times 100 = 63.6V$$

3) Plot the level of V_{o1} , V_{o2} . (6 points)



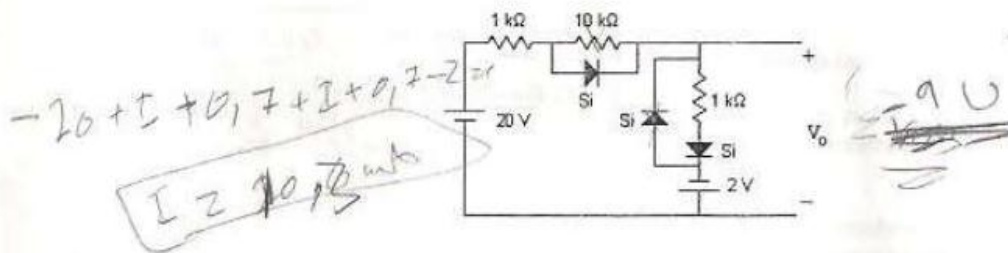
Good Luck



Q.2

(6 marks)

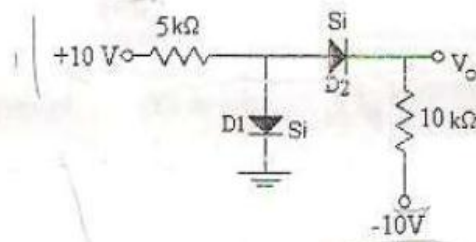
- Determine the output voltage for the circuit below?
- Determine all currents?



Q.3

(5 marks)

Determine I_{D1} and V_o for the circuit below?





Electronic devices

second exam 16/5/2010

Q-1- Given $V_B = 4\text{ V}$ for the network of Fig. 1, determine:

- (a)
- V_E
- . (b)
- I_C
- . (c)
- V_C
- . (d)
- V_{CE}
- . (e)
- I_B
- . (f)
- β
- .

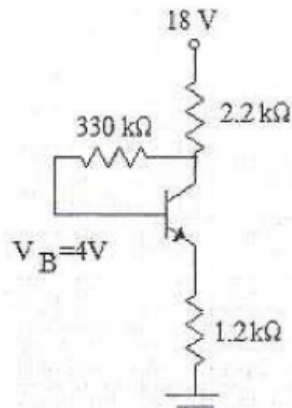


Fig.1

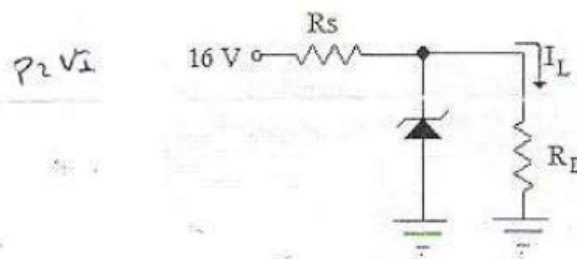
Q-2-(a) Design the network of Fig. 2 to maintain V_L at 12 V for a load variation (I_L) from 0 to 200 mA. That is, determine R_S and V_Z .(b) Determine $P_{Z_{max}}$ for the Zener diode of part (a).

Fig.2

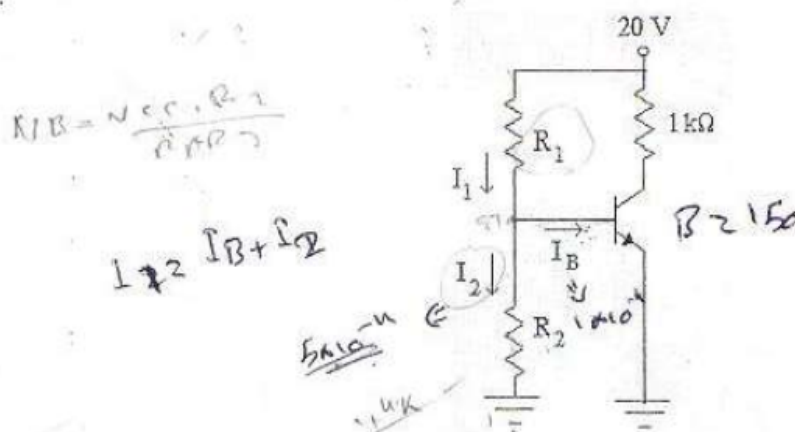
Q-3- Determine R_1 and R_2 for the circuit on fig.3 with the following information $I_1 = 0.6\text{ mA}$, $I_C = 15\text{ mA}$, $\beta = 150$.

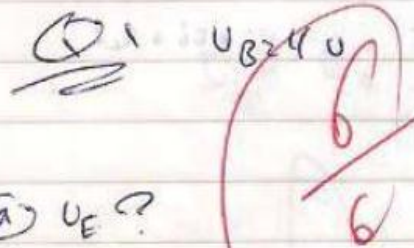
Fig.3

Best wishes for all

DR.majed; DR.hasan; ENG.mahmoud

الرجاء عدم كتابة

في ٢-١



(a) V_E ?

$$V_{BE} = V_B - V_E$$

$$0.7 = 18 - V_E \Rightarrow V_E = 17.3V$$

$$(b) I_C \approx I_E = \frac{V_E}{R_E} = \frac{17.3}{1.2k} = 14.4mA$$

(c) V_C ?

$$-18 + 2.2k \cdot 14.4mA + V_C = 0$$

$$V_C = 11.95V$$

(d) V_{CE} ? $V_{CE} = V_C - V_E$

$$V_{CE} = 11.95 - 17.3 = -5.35V$$

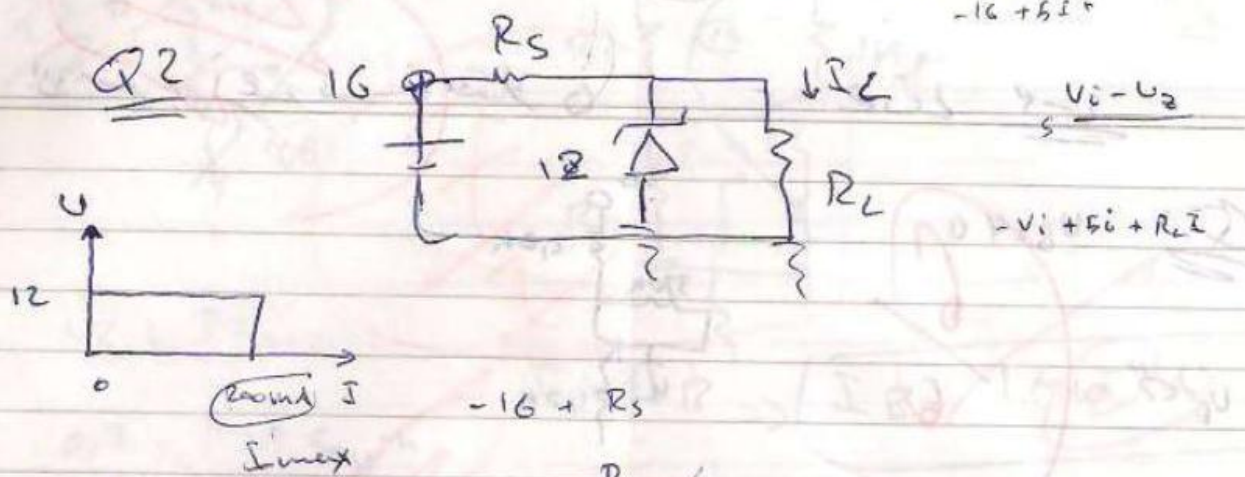
(e) I_B ?

$$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta}$$

$$-18 + 2.2k \cdot I_C + 330k \cdot I_B + 1 = 0$$

$$I_B = 24.1\mu A$$

$$-16 + 5I$$



$$-16 + R_S I_L$$

$$-16 + R$$

$$-16$$

$$I_{max} = \frac{V_Z}{R_{min}}$$

$$-16$$

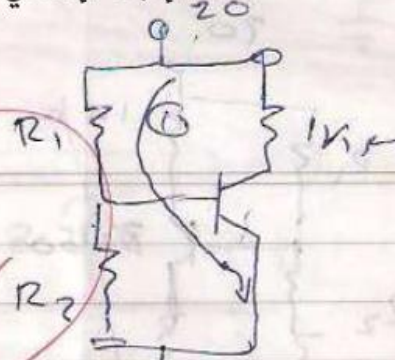
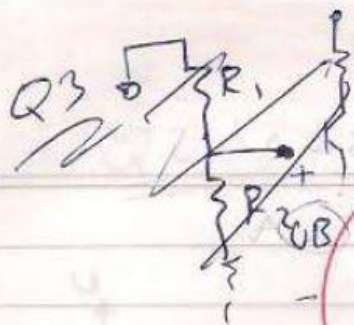
$$I_{max} = \frac{V_Z}{R_{min}}$$

$$200mA = \frac{12}{R_{min}} \Rightarrow R_{min} = 60\Omega$$

(a) $-16 + R_S \times 200mA + 60 \times 200mA$

$R_S \geq 20\Omega$ $V_Z \geq 12$

(b) $P_{max} = 12 \times 200mA = 2.4W$



$$V_B = R_2 I_2$$

$$0.7, R_2 = 5 \times 10^4 \Omega$$

$$R_2 = 1.4 \text{ k}\Omega$$

$$I_B = \beta I_B \Rightarrow$$

$$I_B = 1 \times 10^{-4} \text{ A}$$

$$I_2 = I_B + I_1$$

$$I_2 = 225 \times 10^{-4}$$

$$-20 + R_1 \times 0.6 \text{ mA} + 0.7 \times 20$$

$$R_2 = 232.166 \text{ k}\Omega$$

2

1.26 V

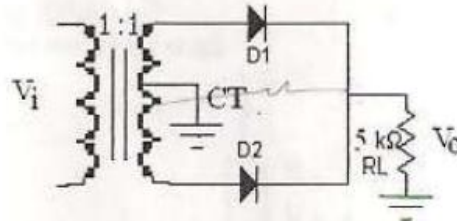


Q.1

(6 marks)

For the circuit in Fig. below, if $V_i = 100 \sin \omega t$ and all diodes are ideal.

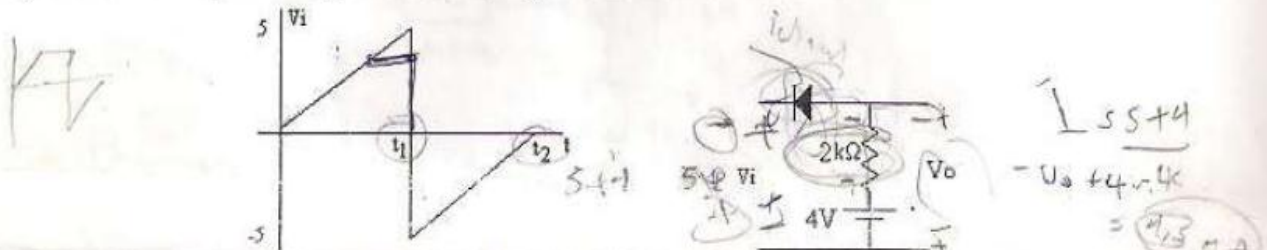
- a) Draw V_o . b) Calculate V_{dc} c) Calculate PIV



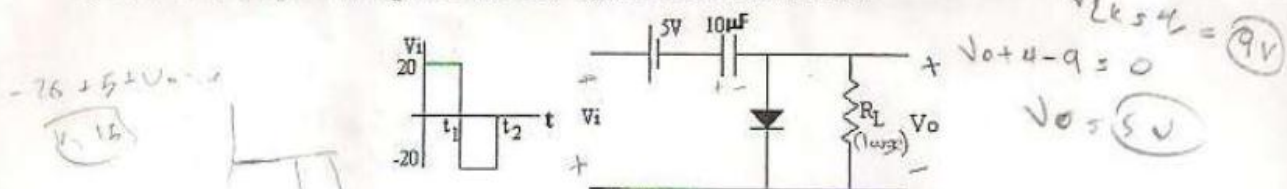
Q.2.

(8 marks)

- a) Draw the output voltage waveform, V_o , for the circuit below



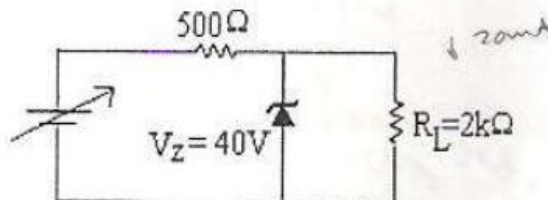
- b) Draw the output voltage waveform, V_o , for the circuit below



Q.3

(6marks)

For the circuit in Fig. below calculate the range of V_i if $I_{Zmin} = 8mA$, $I_{Zmax} = 80mA$ and $V_Z = 40V$



Good Luck

Handwritten calculations for Q.3:

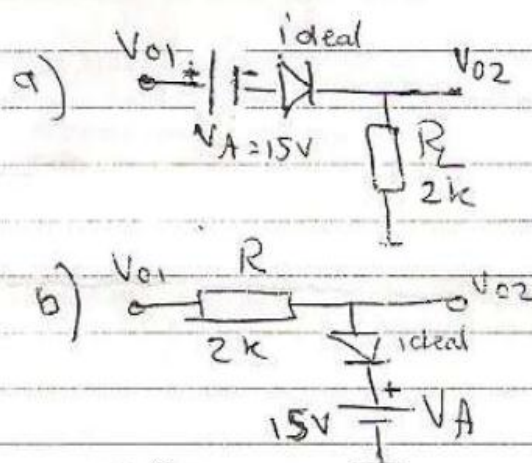
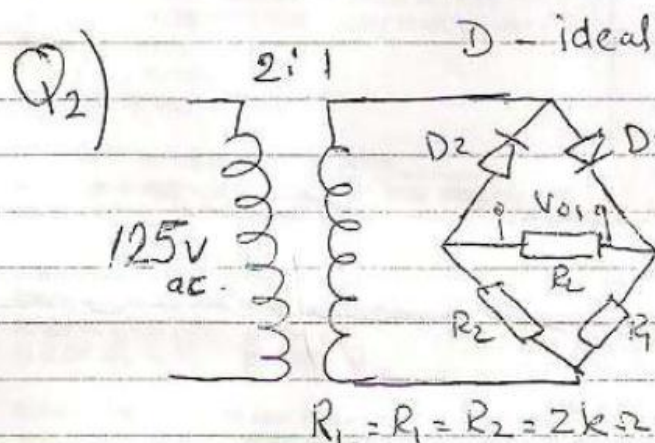
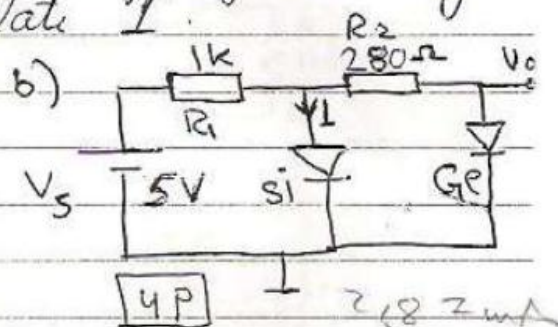
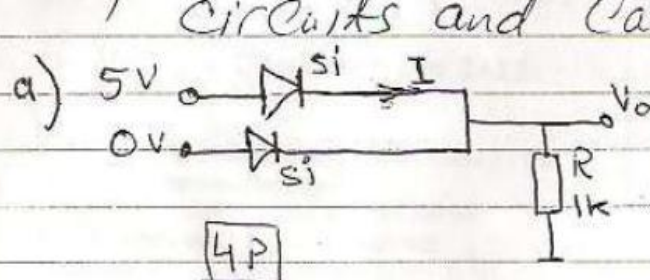
$$V_o = \frac{V_i \cdot 2k}{500 + 2k}$$

$$V_i \approx 750$$

$$V_i \approx 90$$

AL-Balqa Applied University
Faculty of Engineering Technology
Electronic Devices 1st Exam 15.03.07

Q₁) Determine the level of V_o for following circuits and Calculate I .



Using V_{o1} to determine while the limiter is working to charge a battery of 15V dc. choose the circuit which doing that. plot V_{o2}, V_{o1}

12P

good Luck.

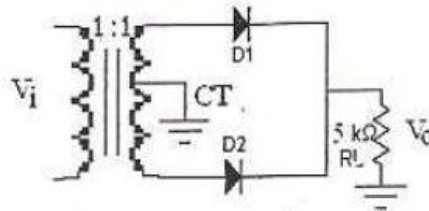


Q.1

(6 marks)

For the circuit in Fig. bellow, if $V_i = 100 \sin \omega t$ and all diodes are ideal.

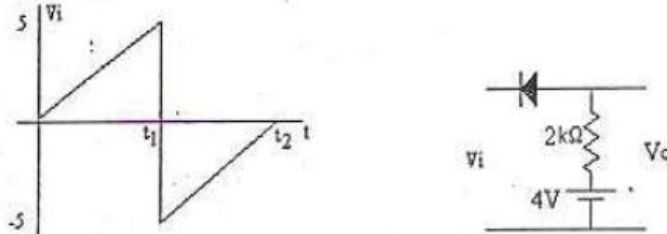
- a) Draw V_o . b) Calculate V_{dc} c) Calculate PIV



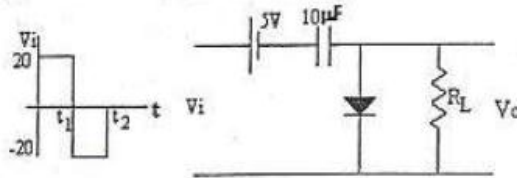
Q.2.

(8 marks)

- a) Draw the output voltage waveform, V_o , for the circuit below



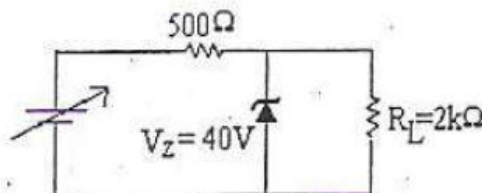
- b) Draw the output voltage waveform, V_o , for the circuit below



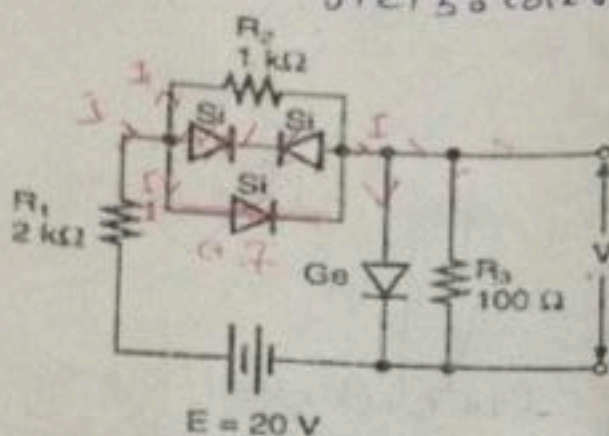
Q.3

(6marks)

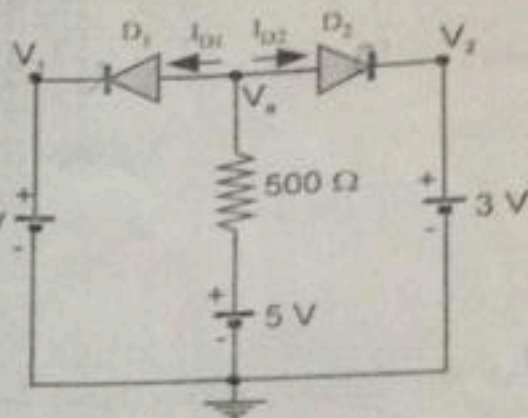
For the circuit in Fig. bellow calculate the range of V_i if $I_{Zmin} = 8mA$, $I_{Zmax} = 80mA$ and $V_Z = 40V$



Good Luck



(Fig.1)



(Fig.2)

Q-1-for the circuit on figure1 answer the following questions: (all diodes are ideal)

- 1-the value of V_0 is:
a) 0.4V b) 0.3V c) 0.7 V d) none of all
- 2 -the value of V_{R2} is:
a) 0 V b) 20 V c) 0.7V d) none of all
- 3- the value of V_{R1} is :
a) 0 V b) 19 V c) 1.4 V d) none of all
- 4-the value of current on R1 is:
a) 9.5 mA b) 0 mA c) 10 mA d) none of all
- 5- the value of the current through R3 is:
a) 4mA b) 3mA c) 0.2A d) none of all

Q-2-for the circuit on figure2 answer the following questions: (all diodes are ideal)

- 6- the value of I_{D1} is:
a) 10mA b) 0 c) 4mA d) none of all
- 7- the value of I_{D2} is:
a) 4mA b) 0 c) 10mA d) none of all
- 8- the value V_{D2} is:
a) 0.7V b) -3V c) 0V d) none of all

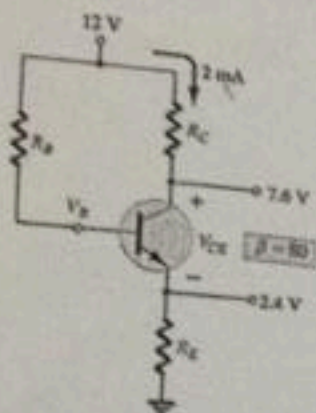


Fig.1

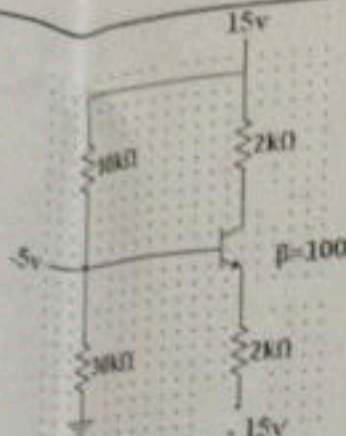


Fig.2

12
20

Q.1) Given the information provided in figure.1 determine:

(6 marks)

- [1] The value of R_C is: 2.2 kΩ
- [2] The value of R_E is: 1.2 kΩ
- [3] The value of R_B is: 3.56 kΩ
- [4] The value of V_B is: 3.1 V

5

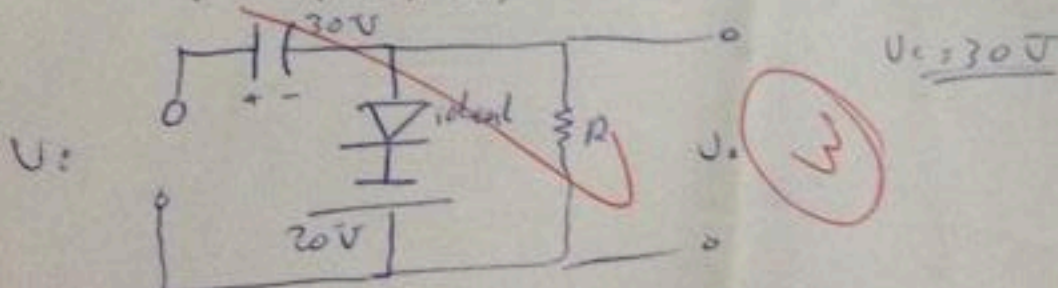
Q.2) For the circuit shown in figure.2 answer the following questions:

(8 marks)

- [1] The value of R_{TH} is: 7.5 kΩ
- [2] The value of V_{TH} is: 2.5 V
- [3] The emitter current is: 8.099 mA
- [4] The value of V_{CE} is: 2.076 V

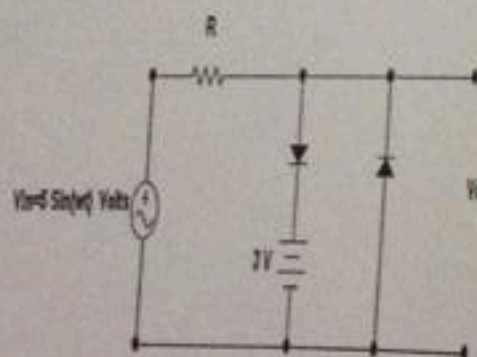
Q.3) Design a clamper that has input with pulse shape (-10V, 10V) and has output pulse shape with (-20V, -40V)

(3 marks)

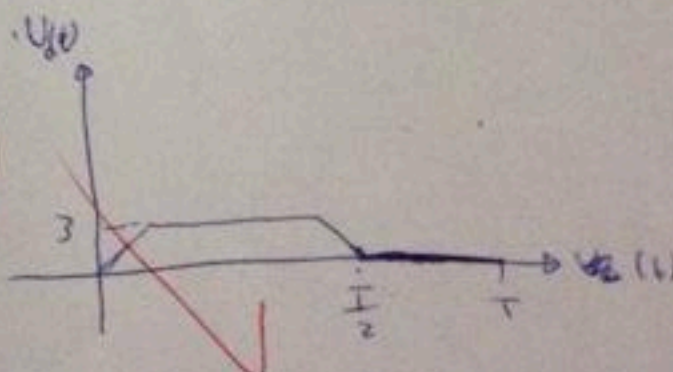


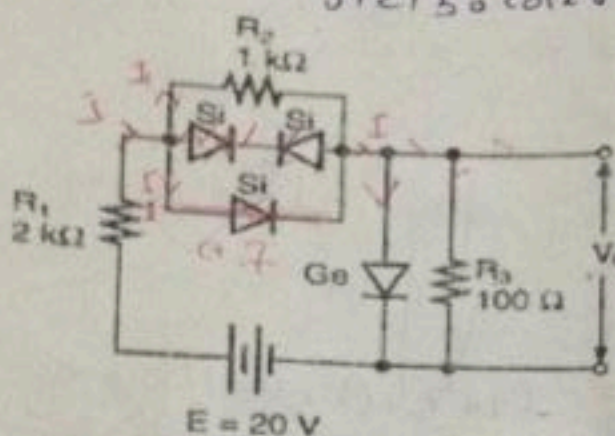
3

Q.4) Draw the output waveform for the figure below, (the diodes are ideals): (3 marks)

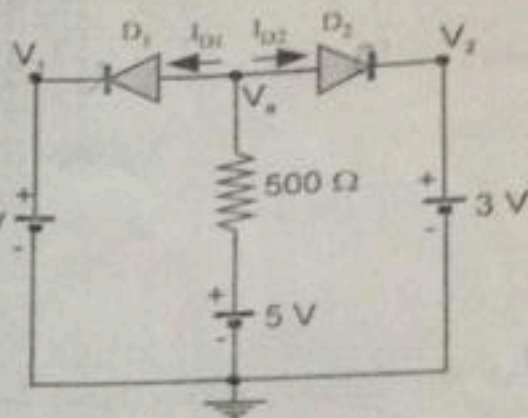


3





(Fig.1)



(Fig.2)

Q-1-for the circuit on figure1 answer the following questions: (all diodes are ideal)

- 1-the value of V_o is:
a) 0.4V b) 0.3V c) 0.7 V d) none of all
- 2 -the value of V_{R2} is:
a) 0 V b) 20 V c) 0.7V d) none of all
- 3- the value of V_{R1} is :
a) 0 V b) 19 V c) 1.4 V d) none of all
- 4-the value of current on R1 is:
a) 9.5 mA b) 0 mA c) 10 mA d) none of all
- 5- the value of the current through R3 is:
a) 4mA b) 3mA c) 0.2A d) none of all

Q-2-for the circuit on figure2 answer the following questions: (all diodes are ideal)

- 6- the value of I_{D1} is:
a) 10mA b) 0 c) 4mA d) none of all
- 7- the value of I_{D2} is:
a) 4mA b) 0 c) 10mA d) none of all
- 8- the value V_{D2} is:
a) 0.7V b) -3V c) 0V d) none of all

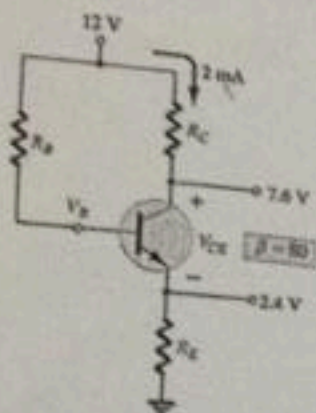


Fig.1

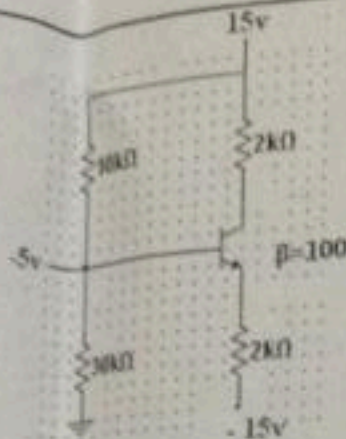


Fig.2

$$\frac{12}{20}$$

Q.1) Given the information provided in figure.1 determine:

(6 marks)

- [1] The value of R_C is: 2.2 kΩ
- [2] The value of R_E is: 1.2 kΩ
- [3] The value of R_B is: 3.56 kΩ
- [4] The value of V_B is: 3.1 V

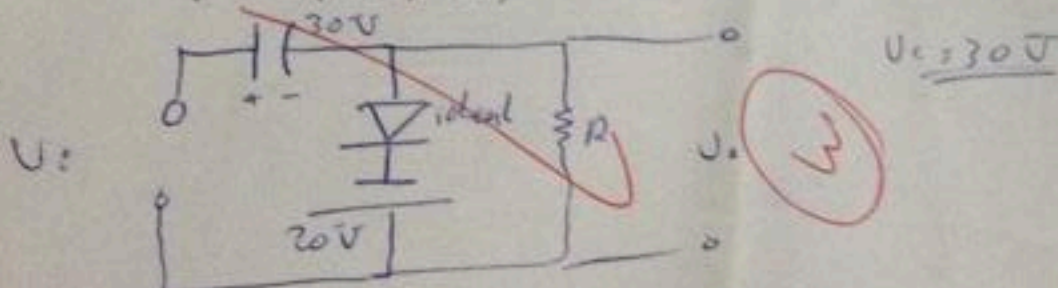
Q.2) For the circuit shown in figure.2 answer the following questions:

(8 marks)

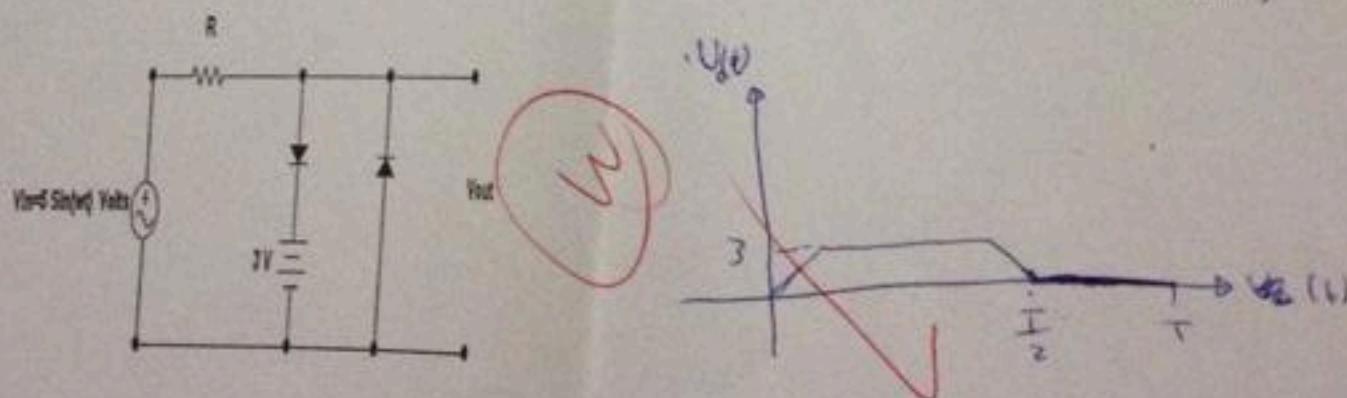
- [1] The value of R_{TH} is: 7.5 kΩ
- [2] The value of V_{TH} is: 2.5 V
- [3] The emitter current is: 8.099 mA
- [4] The value of V_{CE} is: 2.076 V

Q.3) Design a clamper that has input with pulse shape (-10V, 10V) and has output pulse shape with (-20V, -40V)

(3 marks)



Q.4) Draw the output waveform for the figure below, (the diodes are ideals): (3 marks)



Q-2) Answer the following questions about figure 3:

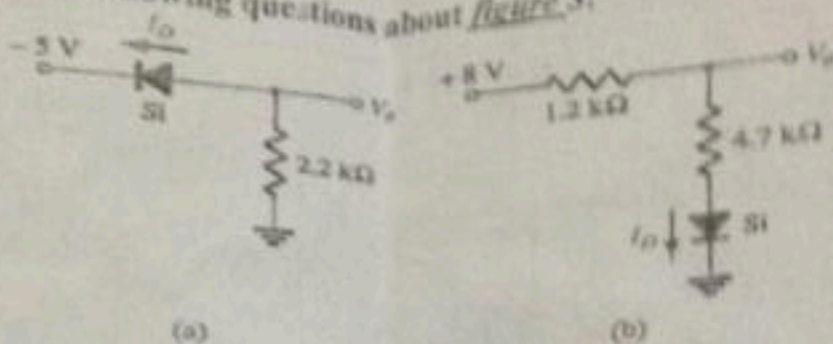


Fig.3

- 1.954 mA) The current I_D on figure 2-(a) is: $\frac{-5 + 0.7 + (2.2 \text{ k}\Omega) I_D}{2.2 \text{ k}\Omega} = 0 \Rightarrow I_D = \frac{4.3 \text{ V}}{2.2 \text{ k}\Omega} = 1.954 \text{ mA}$
- B) The current I_D on figure 2-(b) is: 1.237 mA .
- C) The value of V_o on figure 2-(a) is: -4.3 V .
- D) The value of V_o on figure 2-(b) is: 6.5 V .

No. of questions	1	2	3	4	5	6	7	8
Answer	B	A	B	A	D	B	A	C

$$\begin{aligned} -5 + 0.7 + (2.2 \text{ k}\Omega) I_D &= 0 \\ I_D &= \frac{-4.3 \text{ V}}{2.2 \text{ k}\Omega} \\ &= 1.954 \text{ mA} \end{aligned}$$

[B] $8 - (1.2 \text{ k}\Omega) I_D - (4.7 \text{ k}\Omega) I_D - 0.7 = 0$

$$7.3 = (5.9 \text{ k}\Omega) I_D \Rightarrow I_D = 1.237 \text{ mA}$$

[C] $V_o + (2.2 \text{ k}\Omega) (1.954 \text{ mA}) = 0$

$$\therefore V_o = -4.3 \text{ V}$$

[D] $V_o - (4.7 \text{ k}\Omega \times 1.237 \text{ mA}) - 0.7 = 0$

$$\therefore V_o = 6.5 \text{ V}$$

Q-2) Answer the following questions about figure 3:

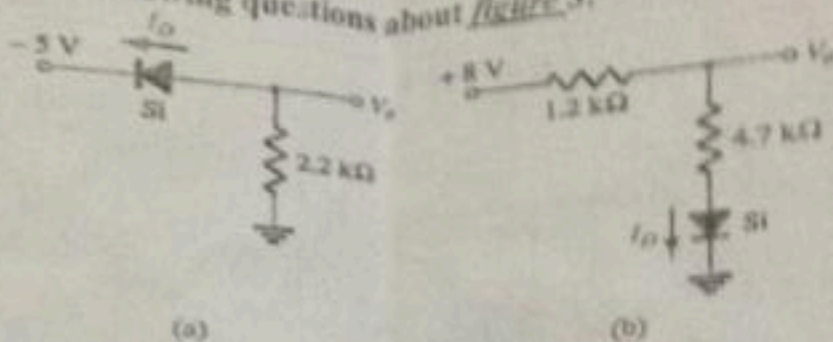


Fig.3

- A) The current I_D on figure 2-(a) is: $\frac{-5 + 0.7 + (2.2 \text{ k}\Omega) I_D}{2.2 \text{ k}\Omega} = 0 \Rightarrow I_D = \frac{4.3 \text{ V}}{2.2 \text{ k}\Omega} = 1.954 \text{ mA}$
- B) The current I_D on figure 2-(b) is: 1.237 mA
- C) The value of V_o on figure 2-(a) is: -4.3 V
- D) The value of V_o on figure 2-(b) is: 6.5 V

No. of questions	1	2	3	4	5	6	7	8
Answer	B	A	B	A	D	B	A	C

$$\begin{aligned} -5 + 0.7 + (2.2 \text{ k}\Omega) I_D &= 0 \\ I_D &= \frac{-4.3 \text{ V}}{2.2 \text{ k}\Omega} \\ &= 1.954 \text{ mA} \end{aligned}$$

[B] $8 - (1.2 \text{ k}\Omega) I_D - (4.7 \text{ k}\Omega) I_D - 0.7 = 0$

$$7.3 = (5.9 \text{ k}\Omega) I_D \Rightarrow I_D = 1.237 \text{ mA}$$

[C] $V_o + (2.2 \text{ k}\Omega) (1.954 \text{ mA}) = 0$

$$\therefore V_o = -4.3 \text{ V}$$

[D] $V_o - (4.7 \text{ k}\Omega \times 1.237 \text{ mA}) - 0.7 = 0$

$$\therefore V_o = 6.5 \text{ V}$$